

$$\begin{bmatrix} 5 & 2 & 3 & 6 \\ -10 & -2 & -5 & -10 \\ -5 & -11 & 1 & 4 \\ 10 & 16 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -11 \\ 20 \\ 7 \\ -14 \end{bmatrix}$$

Setup pivot 1. (nonzero diagonal elements)

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 2 & 0 & 0 & 1 \end{bmatrix}}_{\text{3 shear matrices}} \underbrace{\begin{bmatrix} 5 & 2 & 3 & 6 \\ -10 & -2 & -5 & -10 \\ -5 & -11 & 1 & 4 \\ 10 & 16 & 0 & -2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}}_{\vec{x}} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 2 & 0 & 0 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} -11 \\ 20 \\ 7 \\ -14 \end{bmatrix}}_{\vec{b}}$$

For gauss transform, we take negative of shear matrix values.

Claim:

$$L_1 \cdot A = \begin{bmatrix} * & x & x & x \\ 0 & x & x & x \\ 0 & x & x & x \\ 0 & x & x & x \end{bmatrix}$$

$$\Rightarrow L_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ -2 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{with } L_1 \cdot A = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ -2 & 0 & 0 & 1 \end{bmatrix}}_{L_1} \underbrace{\begin{bmatrix} 5 & 2 & 3 & 6 \\ -10 & -2 & -5 & -10 \\ -5 & -11 & 1 & 4 \\ 10 & 16 & 0 & -2 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 5 & 2 & 3 & 6 \\ 0 & -3 & 1 & 2 \\ 0 & -9 & 4 & 10 \\ 0 & 12 & -6 & -14 \end{bmatrix}}_{L_1 A}$$

Let's find L_2

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -3 & 1 & 0 \\ 0 & 4 & 0 & 1 \end{bmatrix}}_{L_2} \underbrace{\begin{bmatrix} 5 & 2 & 3 & 6 \\ 0 & -3 & 1 & 2 \\ 0 & -9 & 4 & 10 \\ 0 & 12 & -6 & -14 \end{bmatrix}}_{L_1 A} = \underbrace{\begin{bmatrix} 5 & 2 & 3 & 6 \\ 0 & -3 & 1 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -2 & -6 \end{bmatrix}}_{L_2 L_1 A}$$

Recall that Gauss transform is a rank-one update.

$$L_2 = I_4 - \vec{t}_2 \vec{e}_2^T \quad \text{where } \vec{t} = \begin{bmatrix} 0 \\ 0 \\ 3 \\ -4 \end{bmatrix}$$

Moving on to pivot 3.

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 1 \end{bmatrix}}_{L_3} \underbrace{\begin{bmatrix} 5 & 2 & 3 & 6 \\ 0 & -3 & 1 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -2 & -6 \end{bmatrix}}_{L_2 L_1 A} = \underbrace{\begin{bmatrix} 5 & 2 & 3 & 6 \\ 0 & -3 & 1 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 2 \end{bmatrix}}_U$$

$$\text{where } L_3 = I_4 - \vec{t}_3 \vec{e}_3^T \quad \text{w/ } \vec{t} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -2 \end{bmatrix}$$

Thus, we have

$$L_3 \cdot L_2 \cdot L_1 \cdot A = U$$

and L_k is invertible with

$$L_k = I_n - \vec{t}_k \vec{e}_k^T$$

We note the inverse is quite elegant:

$$\begin{aligned} L_k^{-1} &= (I_n - \vec{t}_k \vec{e}_k^T)^{-1} \\ &= I_n + \vec{t}_k \vec{e}_k^T \end{aligned}$$

$$\text{Proof. } L_k \cdot L_k^{-1} = (I_n - \vec{t}_k \vec{e}_k^T)(I_n + \vec{t}_k \vec{e}_k^T)$$

Recall: i. $A \cdot (B+C) = A \cdot B + A \cdot C$
ii. $(A+B) \cdot C = A \cdot C + B \cdot C$

$$\begin{aligned} &= (I_n - \vec{t}_k \vec{e}_k^T) I_n + (I_n - \vec{t}_k \vec{e}_k^T) \vec{t}_k \vec{e}_k^T \\ &= I_n - \underbrace{\vec{t}_k \vec{e}_k^T + \vec{t}_k \vec{e}_k^T}_0 - \vec{t}_k \vec{e}_k^T \vec{t}_k \vec{e}_k^T \\ &= I_n - \vec{t}_k (\underbrace{\vec{e}_k^T \vec{t}_k}_0) \vec{e}_k^T \quad \text{Note: } \vec{t}_k = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ t_{k+1} \\ \vdots \\ t_n \end{bmatrix} \leftarrow k\text{th entry} \\ &= I_n \end{aligned}$$

$$\text{Entry } (\vec{e}_k^T \vec{t}_k)_{kk} = k \cdot 0 = 0$$

$$\Rightarrow L_1^{-1} L_2^{-1} L_3^{-1} L_3 L_2 L_1 A = L_1^{-1} L_2^{-1} L_3^{-1} U$$

$$\Rightarrow A = L_1^{-1} L_2^{-1} L_3^{-1} U$$

Now, let's find L^{-1} .

$$L_1^{-1} \cdot L_2^{-1} = (I_n + \vec{t}_1 \vec{e}_1^T) \cdot (I_n + \vec{t}_2 \vec{e}_2^T)$$

$$= I_n + \vec{t}_2 \vec{e}_2^T + \vec{t}_1 \vec{e}_1^T + \cancel{\vec{t}_1 \vec{e}_1^T \vec{t}_2 \vec{e}_2^T}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & -4 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ -1 & 3 & 1 & 0 \\ 2 & -4 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 5 & 2 & 3 & 6 \\ -10 & -2 & -5 & -10 \\ -5 & -11 & 1 & 4 \\ 10 & 16 & 0 & -2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ -1 & 3 & 1 & 0 \\ 2 & -4 & 0 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 5 & 2 & 3 & 6 \\ 0 & -3 & 1 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 2 \end{bmatrix}}_U = L U$$

$$A \vec{x} = \vec{b}$$

$$\Rightarrow (L \cdot U) \vec{x} = \vec{b}$$

$$\Rightarrow L \cdot (U \cdot \vec{x}) = \vec{b}$$

$$\Rightarrow L \cdot \vec{y} = \vec{b} \quad \text{and } U \vec{x} = \vec{y}$$

Let's find $\vec{y}$ first.

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ -1 & 3 & 1 & 0 \\ 2 & -4 & 0 & 1 \end{bmatrix}}_L \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} -11 \\ 20 \\ 7 \\ -14 \end{bmatrix}$$

$$\Rightarrow 1 \cdot y_1 = -11$$

$$\Rightarrow y_1 = -11$$

$$\Rightarrow -2 \cdot y_1 + 1 \cdot y_2 = 20$$

$$y_2 = 20 + 2y_1$$

$$= 20 + 2(-11)$$

$$= 20 - 22$$

$$\Rightarrow y_2 = -2$$

to be continued.